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| 1 | (a) | (i) | The tension force must be greater than the weight so that the resultant force provides a **centripetal force** for the ball to moving along the **arc of a circle**. | [1] |
| | (a) | (ii) | From Fig 1.2, velocity of the ball just before it hits the wall, v = 2.6 m $s^{-1}$<br>By Newton's 2nd Law, $F_{net} = ma_c$<br>Tension force – weight = $mv^2/r$<br>Tension force = $mv^2/r + mg$ = 350 ($2.6^2$ / 5.8 + 9.81)<br>Tension force = 3840 = 3800 N (2 or 3 s.f.) | [1]<br>[1] |
| | (b) | | From the graph, the average force exerted by the wall on the ball<br>$\langle F \rangle = \frac{\Delta p}{\Delta t} = \frac{m\Delta v}{\Delta t} = \frac{350 \times (0.000 - 2.600)}{(1.35 - 1.20)}$<br>$= -6070$ N<br>By Newton's Third Law,<br>(average force by ball on the wall) = - (average force by wall on the ball)<br>= 6070 = 6100 N (2 or 3 s.f.) | [1]<br>[1] |
| | (c) | | The principle of conservation of linear momentum states that the total momentum remains constant provided there is no **net** external force acting on the system.<br>In this case, external force is acting on the wall as the wall is held by the ground, thus the principle does not apply.<br>OR<br>Considering the system as the ball, the wall and the Earth, the total momentum is constant: momentum loss of ball = momentum gain of wall and Earth. | [1]<br>[1] |
| | (d) | | Thermal energy of ball = 12% of initial KE of ball<br>$mc\Delta T$ = 0.12(½ $mv^2$)<br>$\Delta T$ = 0.12 (½ $v^2$)/c<br>= 0.12 (½ x $2.6^2$) / 450<br>= 9.0 x $10^{-4}$ K or °C | [1]<br>[1] |